

ASD Longitudinal RMD

Stan West

2025-04-17

This Markdown summarizes information regarding the data that will be used for a longitudinal analysis of vocabulary acquisition in autistic children. Here we are interested in vocabulary sizes between 3 & 600 words produced. The numbers reported here differ from our first look because of a couple of reasons:

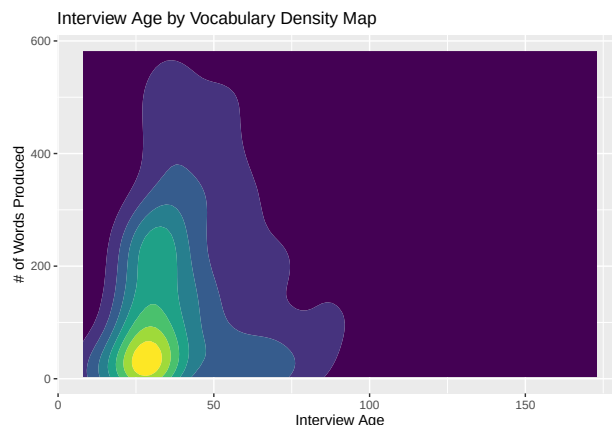
1. I recycled old code from a different project to combine the old and new GUIDs. This code had a part where we were only accepting the “best” CDI record (that with the highest number of words produced across multiple records). This was reducing the number of possible longitudinal data points. Good news, we have more kids with longitudinal data 9 - 15 months between CDIs.
2. I used an “|” instead of “&” when defining the 2 - 4 window for kids with more granular longitudinal data. This made it seem like we had much more kids with fine-tuned longitudinal data. After fixing this, we only have 6 kids with data at that level of specificity.

Demographic Information

Demographics for kids with between 3 & 600 productive vocabulary. The majority of kids have early interviews with relatively low productive vocabularies.

```
## [1] "Number of unique subjectkeys is 350 and the number of unique CDI entries is 629"
```

```
##   subjectkey   interview_age   nProduced
## Length:378524   Min.      :  8.00   Min.      :  3.0
## Class :character 1st Qu.: 27.00   1st Qu.: 32.0
## Mode  :character Median : 36.00   Median :151.0
##                Mean  : 44.34   Mean  :185.7
##                3rd Qu.: 55.00   3rd Qu.:296.0
##                Max.   :173.00   Max.   :582.0
```

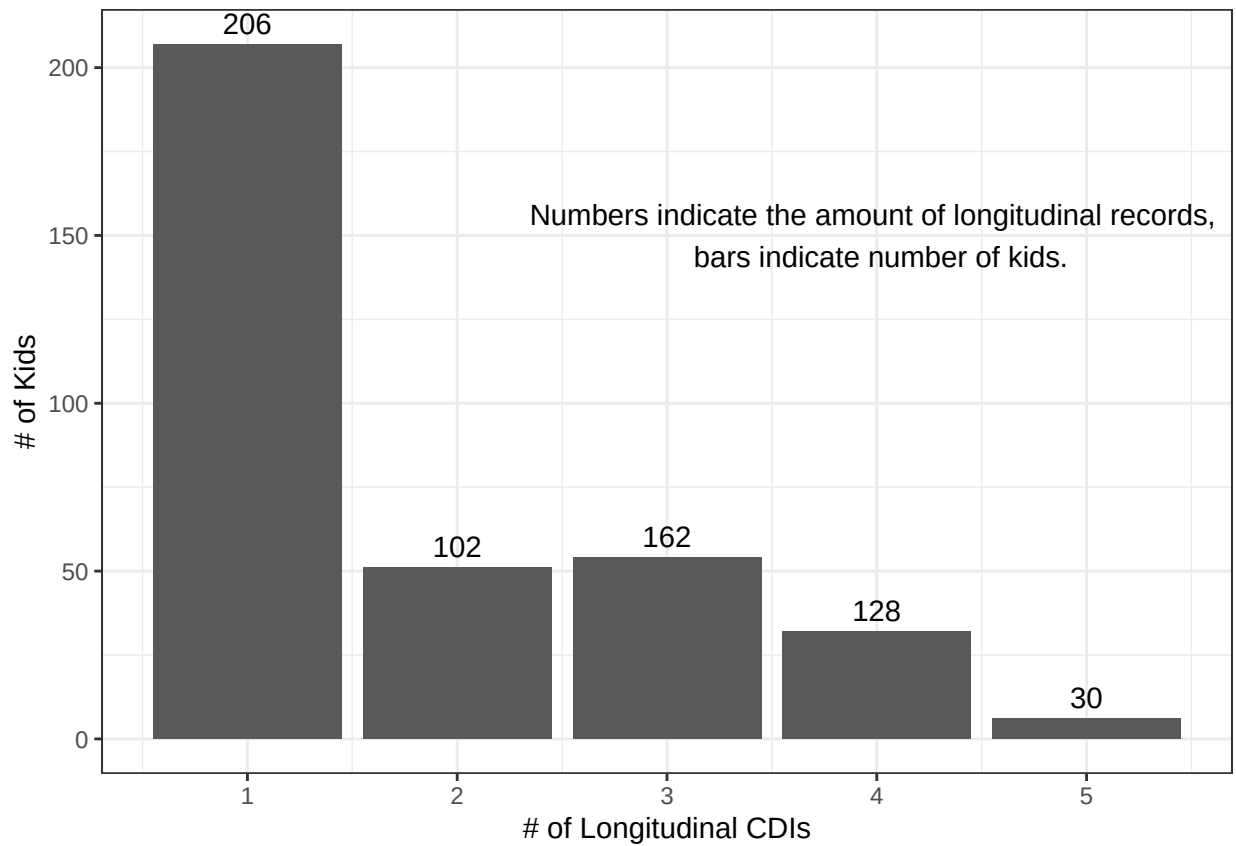


Longitudinal Analysis - all kids (no restriction on time between CDIs)

Of the 628 CDIs, 206 only have an entry for a single age, removing them from longitudinal consideration. That leaves 422(111 unique kids) with at least 2 CDIs.

Table 1: CDI Longitudinal Counts

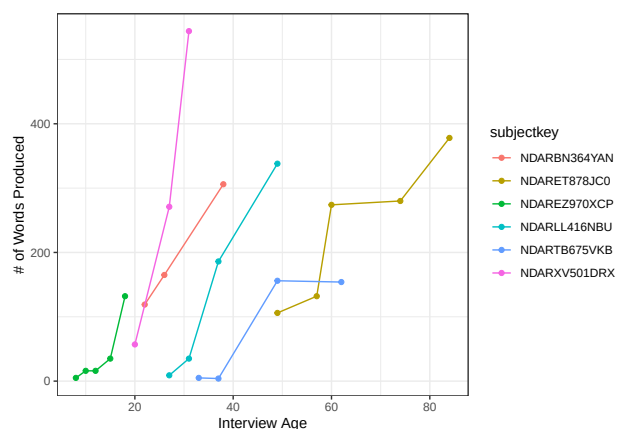
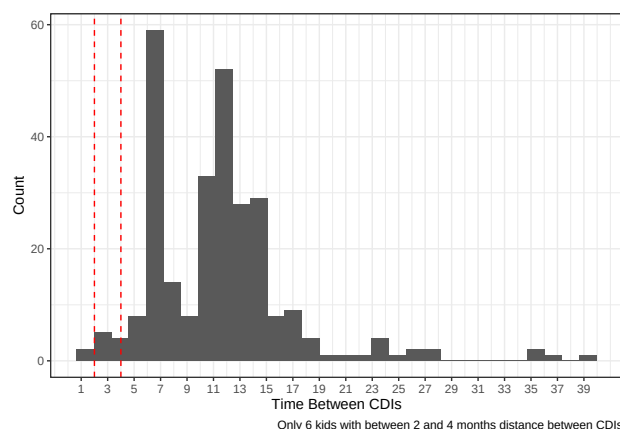
# of CDIs	# of Records	# of Kids
1	207	207
2	102	51
3	162	54
4	128	32
5	30	6



Kids with longitudinal data in 3 month intervals

There are only 6 kids with between 2 and 4 months between CDIs. This is different from our first look at this because we mistakenly used “|” in our filter instead of “&” when defining our time window. There are 9 total records with between 2 and 4 months in between CDIs, with the majority of those (4) being 4 months apart.

subjectkey	interview_age	diff	nProduced
NDARBN364YAN	22	NA	119
NDARBN364YAN	26	4	165
NDARBN364YAN	38	12	306
NDARET878JC0	49	NA	106
NDARET878JC0	57	8	132
NDARET878JC0	60	3	274
NDARET878JC0	74	14	280
NDARET878JC0	84	10	378
NDAREZ970XCP	8	NA	5
NDAREZ970XCP	10	2	16
NDAREZ970XCP	12	2	16
NDAREZ970XCP	15	3	35
NDAREZ970XCP	18	3	132
NDARLL416NBU	27	NA	9
NDARLL416NBU	31	4	35
NDARLL416NBU	37	6	186
NDARLL416NBU	49	12	338
NDARTB675VKB	33	NA	5
NDARTB675VKB	37	4	4
NDARTB675VKB	49	12	156
NDARTB675VKB	62	13	154
NDARXV501DRX	20	NA	57
NDARXV501DRX	27	7	271
NDARXV501DRX	31	4	544



Many of the kids with more frequent CDIs (2 - 4 months apart) have several CDI administrations, however, for most of the kids only 2 timepoints will work for this analysis (e.g., “NDARET878JC0” has 5 CDIs, but only vocabularies at 57 and 60 months would be applicable).

Kids with longitudinal data in 9 - 15 month intervals

There are 111 kids with CDIs at the specificity of 9 - 15 months between CDIs. Below are tables and figures showing the distribution of time between CDIs both across participants and how many individual participants have multiple CDIs within the 9 - 15 time range. The majority of CDIs that meet the 9 - 15 month criteria fall in the 12-month time between CDIs. Additionally, the majority (79 of 111) only have one instance of records that are 9 - 15 months apart.

[1] “There are 111 unique kids with CDIs 9 - 15 months between CDIs” [1] “There are 150 cases of CDIs being between 9 and 15 months apart. The counts are shown below.”

Table 3: CDI Latencies Counts

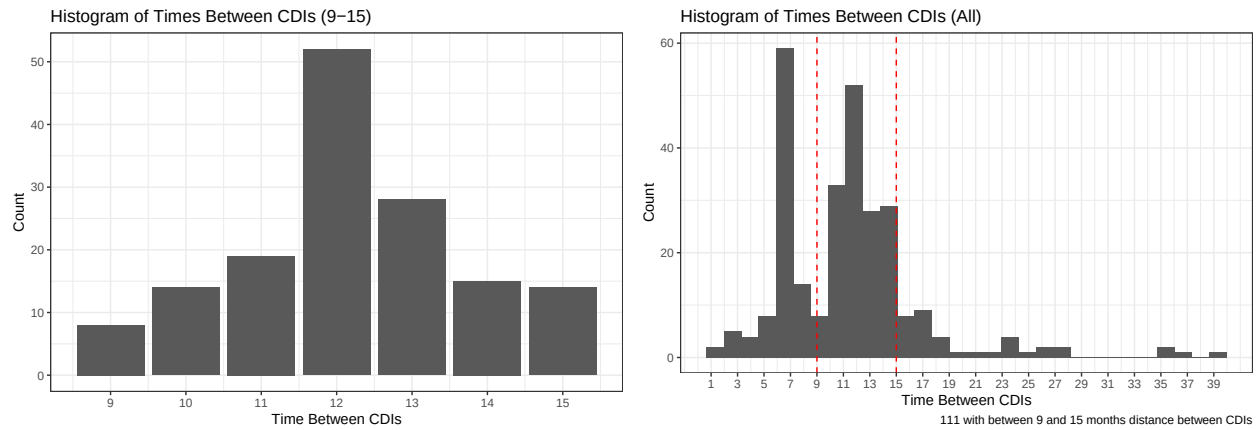
Month Latency	Count
9	8
10	14
11	19
12	52
13	28
14	15
15	14

Table 4: Multiple Contributions Counts: Note: 1 contribution = 2 CDIs 9-15 months apart

# of Contributions	Count
1	79
2	25
3	7

Histograms

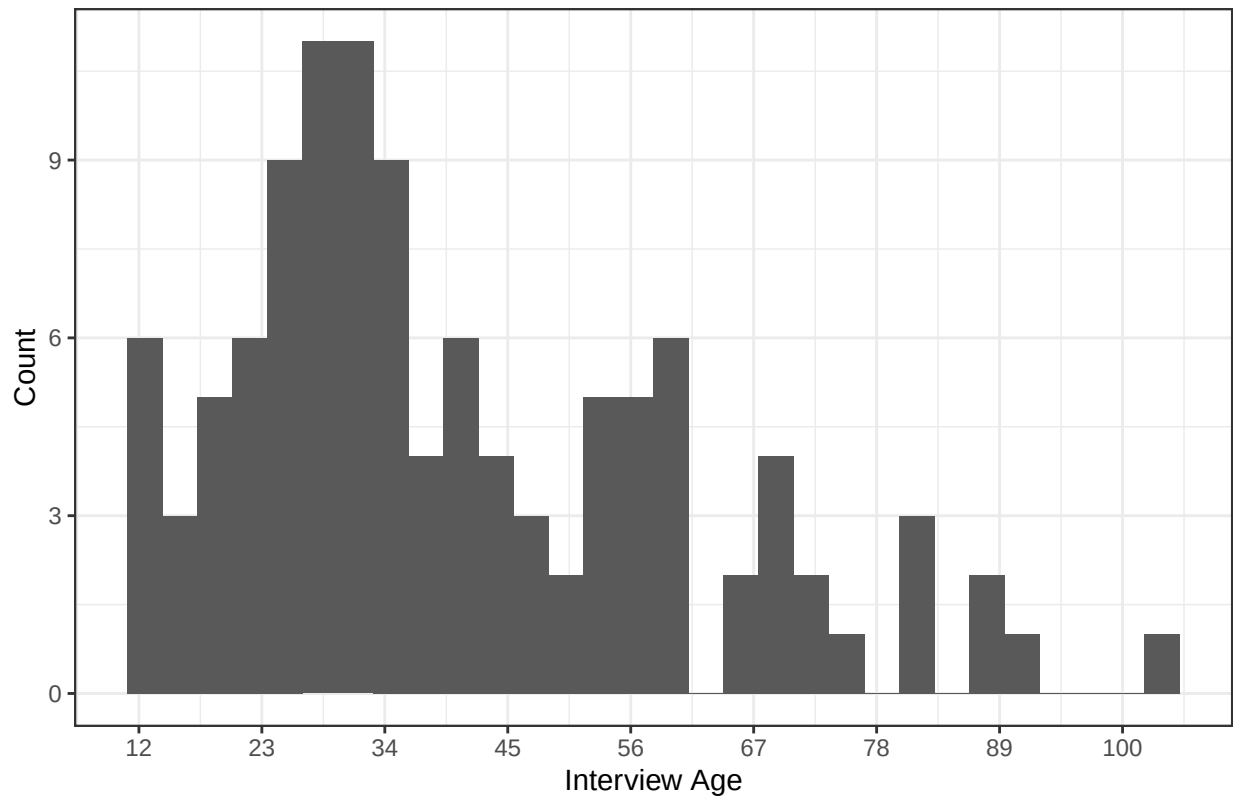
Graphical representation of tables above. Plot on the bottom: red lines reflect the 9 - 15 month cutoffs. If we want to focus on a single time interval between CDIs, we will retain the most data for 12 months between administrations.



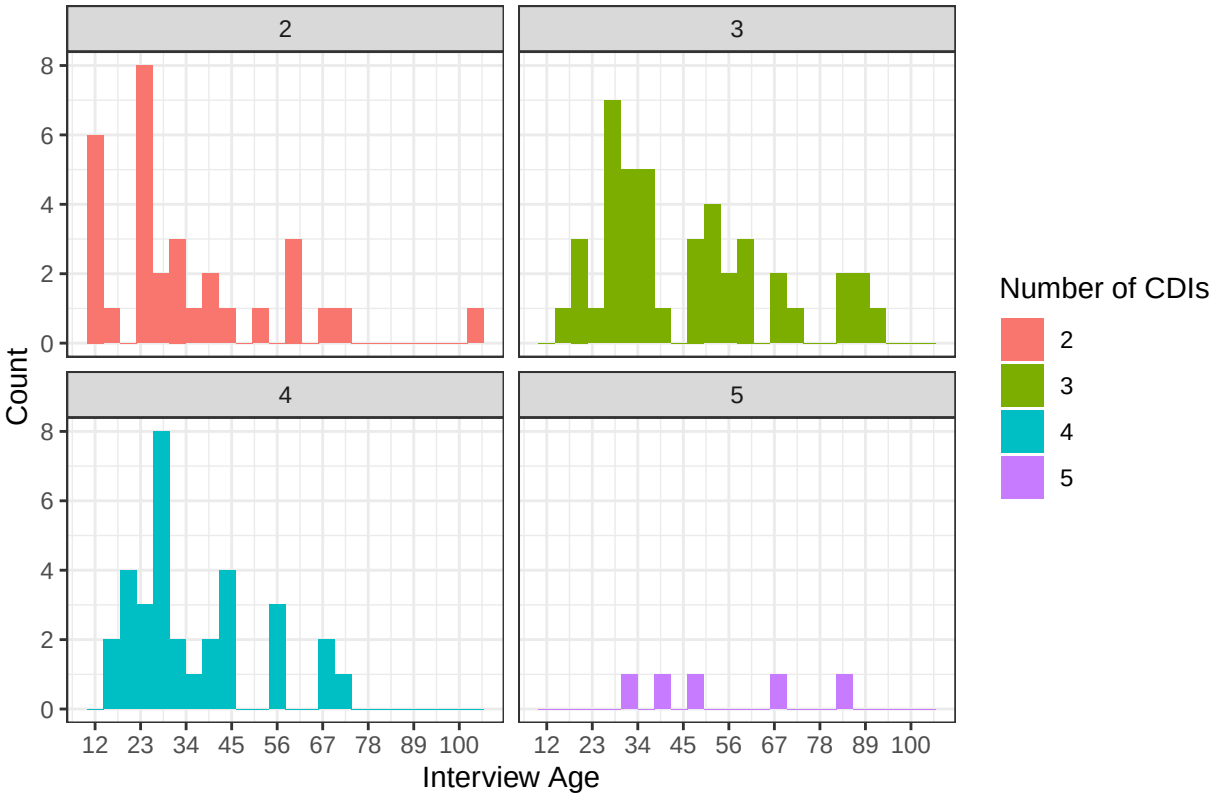
Distributions of first interviews

Histograms show the ages of first administration for kids with longitudinal data between 9 and 15 months apart. The second plot facets this by the number of longitudinal data points available.

Distribution of First Interviews



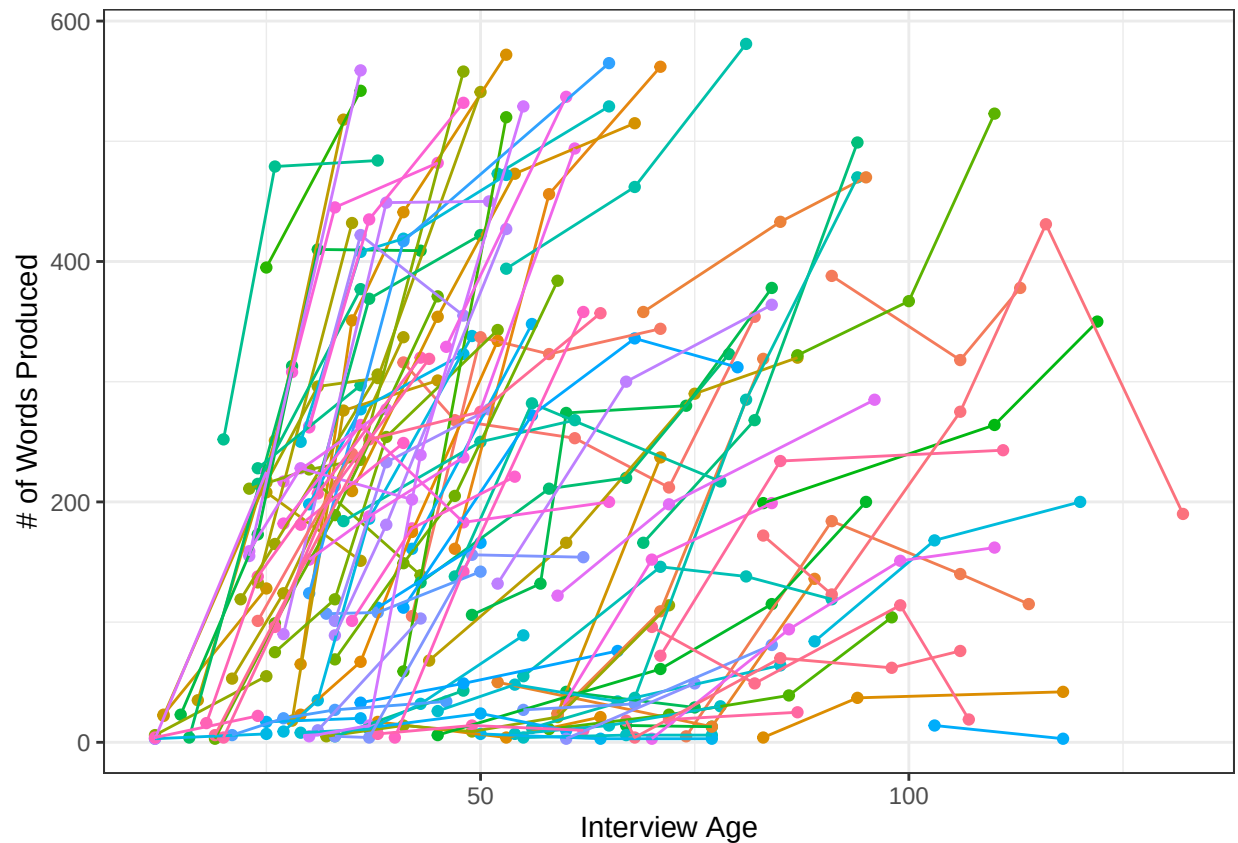
Distribution of First Interviews

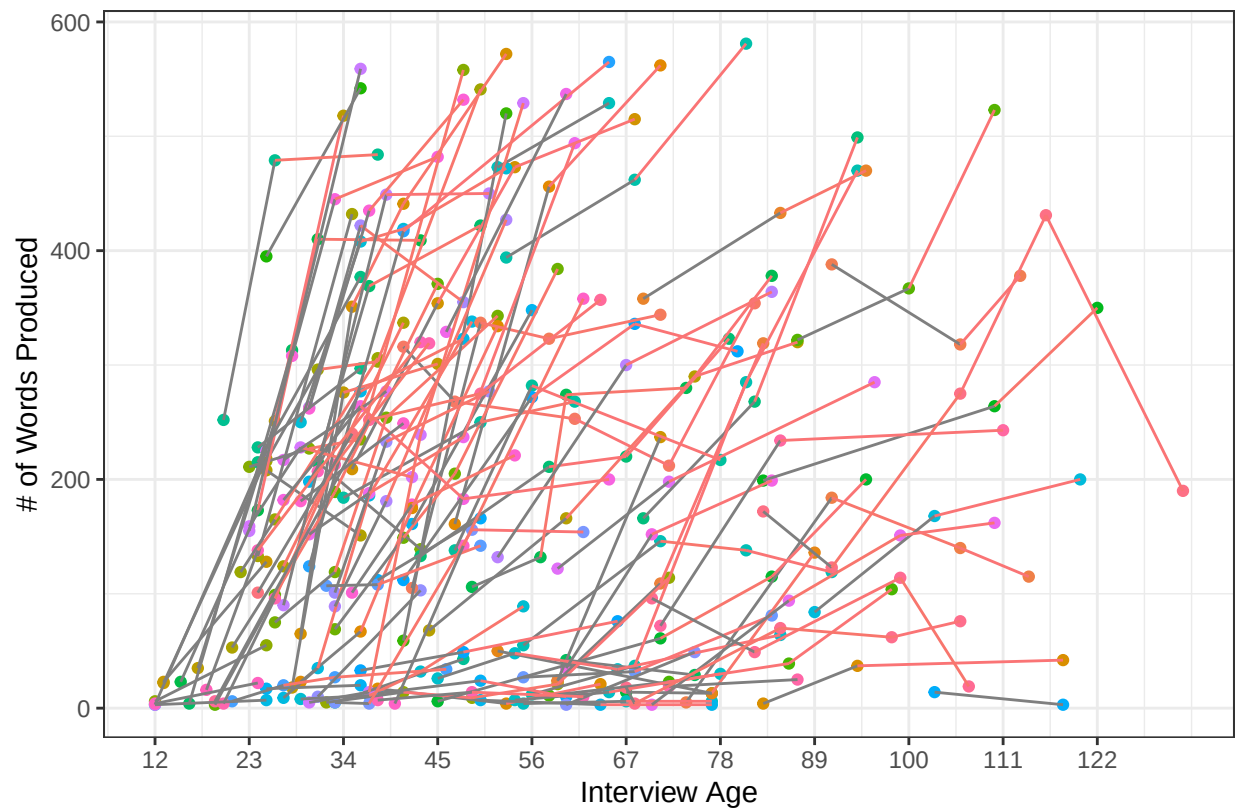


Spaghetti Plots

Plots showing the progression of productive vocabulary over interview age. The majority of kids show increases in productive vocabulary over time, but this is not always the case. Page 10 digs deeper into those cases.

The big dip is subjectkey NDARZV011HWD. It looks like the earlier administration (at 116 months) was words and sentences and the later administration (132 months) was words and gestures. Doesn't seem to be a scoring error: the words previously scored 1 in words and sentences (produces) were later scored 0 in words and gestures (does not produce or understand), so it doesn't seem to be that they mixed up the "understand" and "produces" codes between forms, just that they no longer considered it to be produced.



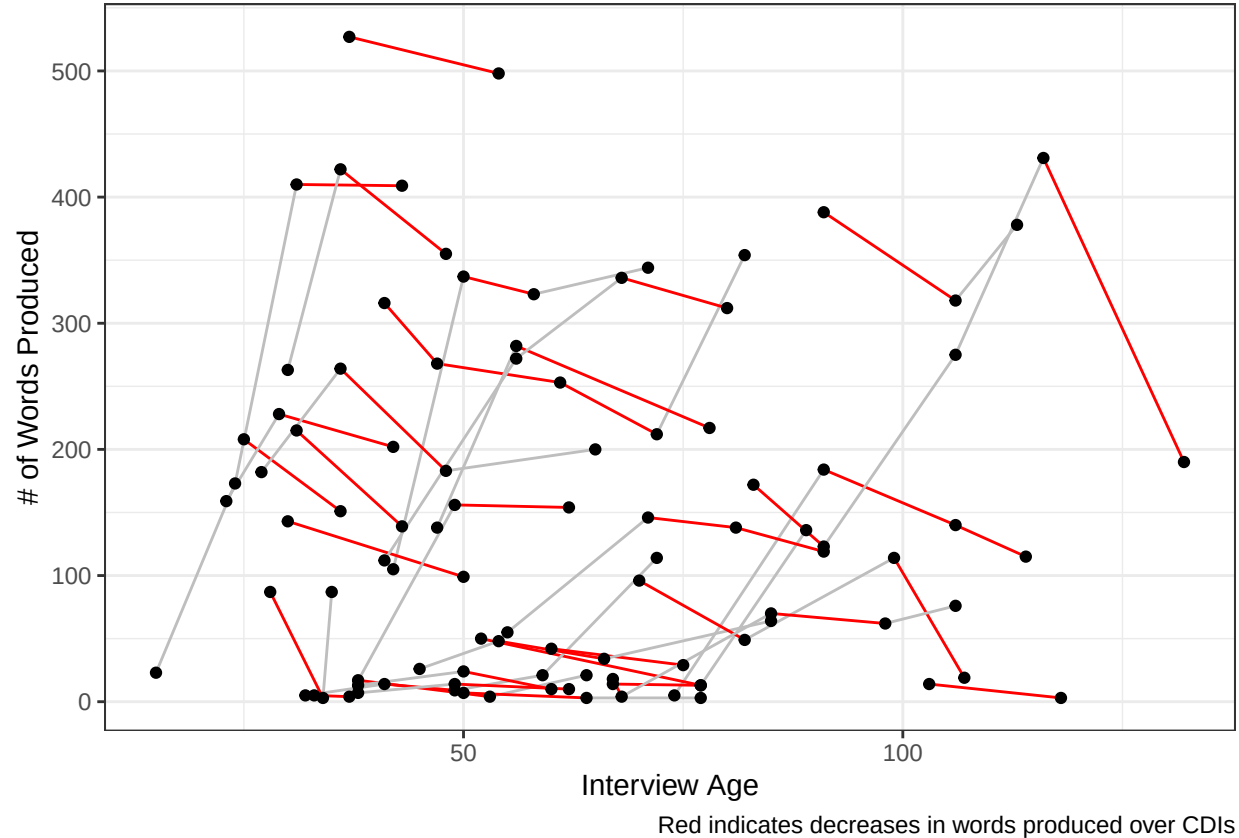


Red connections are 9–15 month gaps between CDIs

Inconsistent gains in vocabulary over time

Some of the kids show drops in vocabulary size between CDI administrations. There are 30 kids who have longitudinal CDIs that decrease at some point in development. These may be considerations for removal as they do not follow the pattern of development of the majority of the kids. It is impossible to know if this is due to bad reporting or legitimate losses in productive vocabulary (Arunachalam, S., Avtushka, V., Luyster, R. J., & Guthrie, W. (2022). Consistency and inconsistency in caregiver reporting of vocabulary. *Language Learning and Development*, 18(1), 81-96. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9119658/pdf/nihms-1714234.pdf>).

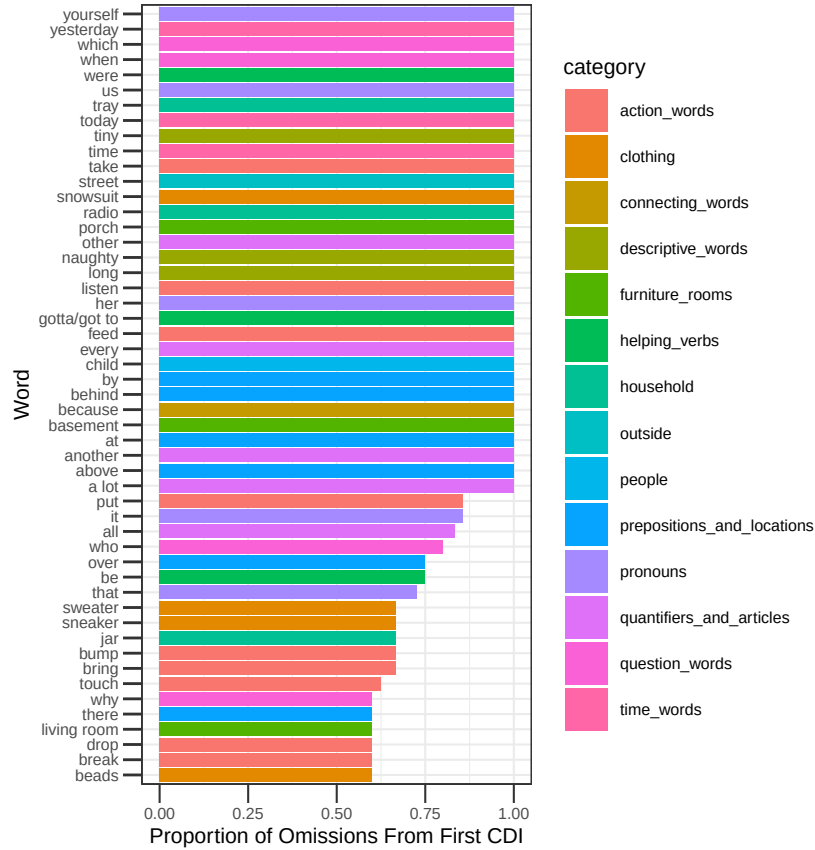
It could be the case that some of the parents or administrators put 0s in the cases that they did not have data for a word (most of the CDI entries just put NA if there is no data for a word), but it is impossible to tell if those 0s actually reflect non-production or inconsistent reporting of null data. We do filter out records with more than 10% NA in item-level reporting to try and curb some of these incomplete records.



Mean Vocabulary Decreases	SD	Range
-37.13158	43.35409	-241
-37.13158	43.35409	-1

Top 40 words not consistently reported across CDIs.

This is data aggregated across kids with longitudinal data with 9 - 15 months between CDIs where the vocabulary size decreased with age



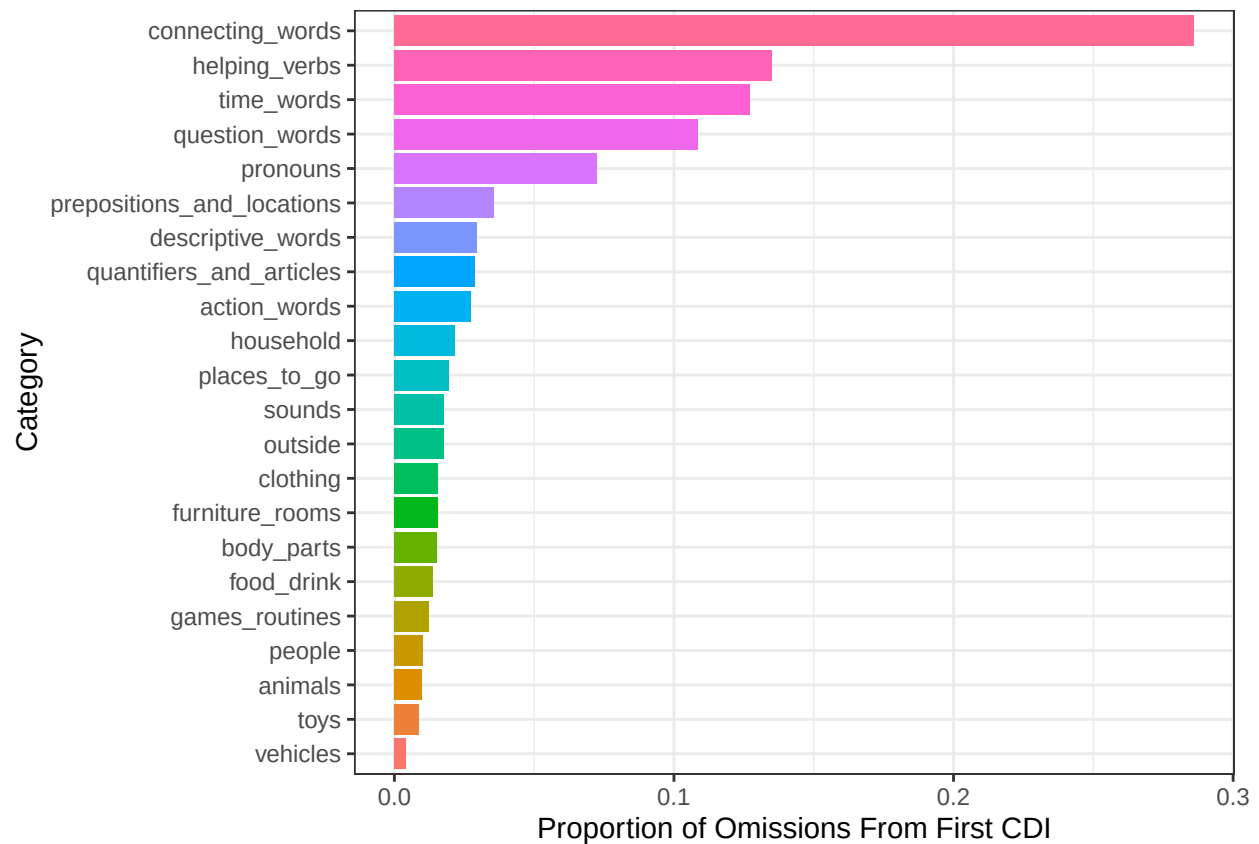
Here, we are looking at item-level distributions of the top 50 words that were reported as being known in the first CDI administration, but not reported as being known in at least one of the subsequent administrations. The bars are filled by CDI category. Values on the x-axis report the proportion of omissions to the total number of times the word was produced on the first CDI administration.

Table 6: Top 50 Inconsistent Words

Word	Count Inconsistent	Count First Produced	Proportion Inconsistent	Category
yourself	1	1	1.0000000	pronouns
yesterday	1	1	1.0000000	time_words
which	1	1	1.0000000	question_words
when	1	1	1.0000000	question_words
were	1	1	1.0000000	helping_verbs
us	1	1	1.0000000	pronouns
tray	1	1	1.0000000	household
today	1	1	1.0000000	time_words
tiny	1	1	1.0000000	descriptive_words
time	2	2	1.0000000	time_words
take	1	1	1.0000000	action_words
street	1	1	1.0000000	outside

Word	Count Inconsistent	Count First Produced	Proportion Inconsistent	Category
snowsuit	1	1	1.0000000	clothing
radio	1	1	1.0000000	household
porch	1	1	1.0000000	furniture_rooms
other	1	1	1.0000000	quantifiers_and_articles
naughty	1	1	1.0000000	descriptive_words
long	3	3	1.0000000	descriptive_words
listen	3	3	1.0000000	action_words
her	5	5	1.0000000	pronouns
gotta/got	2	2	1.0000000	helping_verbs
to				
feed	2	2	1.0000000	action_words
every	2	2	1.0000000	quantifiers_and_articles
child	1	1	1.0000000	people
by	1	1	1.0000000	prepositions_and_locations
behind	2	2	1.0000000	prepositions_and_locations
because	1	1	1.0000000	connecting_words
basement	1	1	1.0000000	furniture_rooms
at	2	2	1.0000000	prepositions_and_locations
another	1	1	1.0000000	quantifiers_and_articles
above	3	3	1.0000000	prepositions_and_locations
a lot	1	1	1.0000000	quantifiers_and_articles
put	6	7	0.8571429	action_words
it	6	7	0.8571429	pronouns
all	5	6	0.8333333	quantifiers_and_articles
who	4	5	0.8000000	question_words
over	3	4	0.7500000	prepositions_and_locations
be	3	4	0.7500000	helping_verbs
that	8	11	0.7272727	pronouns
sweater	4	6	0.6666667	clothing
sneaker	2	3	0.6666667	clothing
jar	2	3	0.6666667	household
bump	4	6	0.6666667	action_words
bring	2	3	0.6666667	action_words
touch	5	8	0.6250000	action_words
why	6	10	0.6000000	question_words
there	6	10	0.6000000	prepositions_and_locations
living room	3	5	0.6000000	furniture_rooms
drop	3	5	0.6000000	action_words
break	3	5	0.6000000	action_words
beads	3	5	0.6000000	clothing

Overall distribution of omissions across categories



This plot is showing the proportion of inconsistencies across words summarized into their CDI categories. First I took the count of the number of total inconsistencies in a category and divided by the total number of times words from that category were produced in the first CDI administration. These proportions were then divided by the sum of those proportions to ascertain the degree to which individual categories were contributing to inconsistent vocabulary reporting.

Skips 12 months apart for more than 2 CDIs

Some kids have CDIs that are around 12 months apart between two administrations, but there is a larger gap in subsequent administrations. We can make predictions about the vocabulary size in these gaps using linear models. Below are some examples of kids with 12 month administrations and larger gaps in times between CDIs. Some kids start with more frequent CDIs and they become less frequent over time (e.g., NDARAJ572PGT) and some start with less frequent CDIs and progressively get more frequent (NDAREH113WHB).

Table 7: Kids with 12 month ‘skips’ in CDI administrations

subjectkey	interview_age	entry	diff	max_diff	n_entry	nProduced
NDARAJ572PGT	83	1	NA	24	3	4
NDARAJ572PGT	94	2	11	24	3	37
NDARAJ572PGT	118	3	24	24	3	42
NDARED843XK0	83	1	NA	27	3	199
NDARED843XK0	110	2	27	27	3	264
NDARED843XK0	122	3	12	27	3	350
NDAREH113WHB	45	1	NA	26	4	6
NDAREH113WHB	71	2	26	26	4	61
NDAREH113WHB	84	3	13	26	4	115
NDAREH113WHB	95	4	11	26	4	200
NDARRC328ERX	30	1	NA	24	3	124
NDARRC328ERX	41	2	11	24	3	417
NDARRC328ERX	65	3	24	24	3	565
NDARVV356TKH	59	1	NA	24	3	122
NDARVV356TKH	72	2	13	24	3	198
NDARVV356TKH	96	3	24	24	3	285
NDARYH321PJE	71	1	NA	26	3	72
NDARYH321PJE	85	2	14	26	3	234
NDARYH321PJE	111	3	26	26	3	243
NDAR_INVMQ312BZS	52	1	NA	25	3	50
NDAR_INVMQ312BZS	77	2	25	25	3	13
NDAR_INVMQ312BZS	89	3	12	25	3	136

[1] "The number of unique kids with 12 month intervals and larger gaps in CDIs is 7"

Skips 6 months apart for more than 2 CDIs

Some kids have CDIs that are around 6 months apart between two administrations, but there is a larger gap in subsequent administrations. We can make predictions about the vocabulary size in these gaps using linear models. Below are some examples of kids with 6 month administrations and larger gaps in times between CDIs.

Table 8: Kids with 6 month ‘skips’ in CDI administrations

subjectkey	interview__age	entry	diff	max_diff	n_entry	nProduced
NDARAK656UY4	29	1	NA	12	4	65
NDARAK656UY4	35	2	6	12	4	351
NDARAK656UY4	41	3	6	12	4	441
NDARAK656UY4	53	4	12	12	4	572
NDARAT218HWY	28	1	NA	11	3	18
NDARAT218HWY	34	2	6	11	3	276
NDARAT218HWY	45	3	11	11	3	301
NDARBE387XAU	21	1	NA	14	3	53
NDARBE387XAU	27	2	6	14	3	124
NDARBE387XAU	41	3	14	14	3	337
NDARBK488WYL	21	1	NA	7	3	11
NDARBK488WYL	27	2	6	7	3	129
NDARBK488WYL	34	3	7	7	3	259
NDARBK744FJ1	25	1	NA	12	4	216
NDARBK744FJ1	31	2	6	12	4	296
NDARBK744FJ1	38	3	7	12	4	303
NDARBK744FJ1	50	4	12	12	4	541
NDARCJ346ALW	23	1	NA	12	4	211
NDARCJ346ALW	30	2	7	12	4	227
NDARCJ346ALW	36	3	6	12	4	235
NDARCJ346ALW	48	4	12	12	4	558
NDARCN229HL4	26	1	NA	13	4	75
NDARCN229HL4	33	2	7	13	4	119
NDARCN229HL4	39	3	6	13	4	254
NDARCN229HL4	52	4	13	13	4	343
NDARCY695ZGF	33	1	NA	12	4	69
NDARCY695ZGF	41	2	8	12	4	149
NDARCY695ZGF	47	3	6	12	4	205
NDARCY695ZGF	59	4	12	12	4	384
NDAREY594LDB	31	1	NA	13	3	215
NDAREY594LDB	37	2	6	13	3	369
NDAREY594LDB	50	3	13	13	3	422
NDARGA086NM5	27	1	NA	6	3	26
NDARGA086NM5	33	2	6	6	3	209
NDARGA086NM5	39	3	6	6	3	323
NDARGJ326MNV	20	1	NA	12	3	252
NDARGJ326MNV	26	2	6	12	3	479
NDARGJ326MNV	38	3	12	12	3	484
NDARJM897HH8	28	1	NA	6	3	87
NDARJM897HH8	34	2	6	6	3	3
NDARJM897HH8	35	3	1	6	3	87
NDARKE927LAD	29	1	NA	12	4	8
NDARKE927LAD	37	2	8	12	4	11

subjectkey	interview_age	entry	diff	max_diff	n_entry	nProduced
NDARKE927LAD	43	3	6	12	4	32
NDARKE927LAD	55	4	12	12	4	89
NDARLL416NBU	27	1	NA	12	4	9
NDARLL416NBU	31	2	4	12	4	35
NDARLL416NBU	37	3	6	12	4	186
NDARLL416NBU	49	4	12	12	4	338
NDARLV040PD8	30	1	NA	12	3	198
NDARLV040PD8	36	2	6	12	3	277
NDARLV040PD8	48	3	12	12	3	323
NDARRE163EMK	21	1	NA	13	4	6
NDARRE163EMK	27	2	6	13	4	20
NDARRE163EMK	33	3	6	13	4	27
NDARRE163EMK	46	4	13	13	4	34
NDARRX042BGL	32	1	NA	12	3	107
NDARRX042BGL	38	2	6	12	3	108
NDARRX042BGL	50	3	12	12	3	142
NDARTP065WRK	33	1	NA	12	3	101
NDARTP065WRK	39	2	6	12	3	233
NDARTP065WRK	51	3	12	12	3	277
NDARTU179NLD	30	1	NA	12	3	263
NDARTU179NLD	36	2	6	12	3	422
NDARTU179NLD	48	3	12	12	3	355
NDARUG689YRF	33	1	NA	14	3	89
NDARUG689YRF	39	2	6	14	3	181
NDARUG689YRF	53	3	14	14	3	427
NDARVH166ABT	30	1	NA	12	4	5
NDARVH166ABT	37	2	7	12	4	14
NDARVH166ABT	43	3	6	12	4	239
NDARVH166ABT	55	4	12	12	4	529
NDARVL353XDE	23	1	NA	13	3	159
NDARVL353XDE	29	2	6	13	3	228
NDARVL353XDE	42	3	13	13	3	202
NDARXU289BV6	18	1	NA	12	4	16
NDARXU289BV6	24	2	6	12	4	138
NDARXU289BV6	31	3	7	12	4	207
NDARXU289BV6	43	4	12	12	4	320
NDARYA737JZ1	24	1	NA	7	3	41
NDARYA737JZ1	31	2	7	7	3	261
NDARYA737JZ1	37	3	6	7	3	389
NDARYA964GGT	20	1	NA	11	4	4
NDARYA964GGT	26	2	6	11	4	96
NDARYA964GGT	33	3	7	11	4	213
NDARYA964GGT	44	4	11	11	4	319
NDARZU928VZZ	22	1	NA	6	3	253
NDARZU928VZZ	27	2	5	6	3	465
NDARZU928VZZ	33	3	6	6	3	509
NDARZV785BAB	25	1	NA	6	3	79
NDARZV785BAB	31	2	6	6	3	479
NDARZV785BAB	37	3	6	6	3	576
NDAR_INVAC323FPZ	41	1	NA	14	5	316
NDAR_INVAC323FPZ	47	2	6	14	5	268
NDAR_INVAC323FPZ	61	3	14	14	5	253

subjectkey	interview_age	entry	diff	max_diff	n_entry	nProduced
NDAR_INVAC323FPZ	72	4	11	14	5	212
NDAR_INVAC323FPZ	82	5	10	14	5	354
NDAR_INVSE125EEG	29	1	NA	10	4	23
NDAR_INVSE125EEG	36	2	7	10	4	67
NDAR_INVSE125EEG	42	3	6	10	4	175
NDAR_INVSE125EEG	52	4	10	10	4	334

[1] "The number of unique kids with 6 month intervals in CDIs and larger gaps is 29"